Modeling with partial differential equations
Part II: the Poisson equation

Tony Lelièvre

Ecole Nationale des Ponts et Chaussées - Institut Polytechnique de Paris



21st September 2026

Definition of solutions to the Poisson equation

In this part, we will be interested in solutions to the following problem: find $u : \mathbb{R}^d \to \mathbb{R}$ such that

$$-\Delta u = f \tag{1}$$

where

$$\Delta u = \sum_{i=1}^{d} \frac{\partial^2 u}{\partial x_i^2} = \sum_{i=1}^{d} \partial_{x_i, x_i} u$$

We consider classical solutions: $f : \mathbb{R}^d \to \mathbb{R}$ is continuous, and $u : \mathbb{R}^d \to \mathbb{R}$ is of class $\mathcal{C}^2$

When $f \neq 0$, Eq. (1) is the Poisson equation
When $f = 0$, Eq. (1) is the Laplace equation, and u is a harmonic function

Remark: the problem is linear. If $-\Delta u_1 = f_1$ and $-\Delta u_2 = f_2$, then, for any $\lambda \in \mathbb{R}$, a solution to $-\Delta u = f_1 + \lambda f_2$ is $u = u_1 + \lambda u_2$

Outline

- Derivation of Laplace and Poisson equations, and some applications
- The Poisson equation in $\mathbb{R}^d$
- The Dirichlet problem in Ω, bounded open subset of $\mathbb{R}^d$
- Numerical approximations using finite differences

Preliminaries: gradient, divergence, and laplacian

We will need some differential operators

For any differentiable function $u : \mathbb{R}^d \to \mathbb{R}$, the gradient of u is the vector field $\nabla u : \mathbb{R}^d \to \mathbb{R}^d$ with, for any $i \in \{1, \ldots, d\}$,

$$[\nabla u]_i = \frac{\partial u}{\partial x_i} = \partial_{x_i} u$$

For any differentiable vector field $F : \mathbb{R}^d \to \mathbb{R}^d$, the divergence of F is the function $\operatorname{div} F : \mathbb{R}^d \to \mathbb{R}$ defined by:

$$\operatorname{div}(F) = \sum_{i=1}^{d} \frac{\partial F_i}{\partial x_i} = \sum_{i=1}^{d} \partial_{x_i} F_i$$

Fundamental result: for any $u : \mathbb{R}^d \to \mathbb{R}$ in $\mathcal{C}^2$, the laplacian of u is

$$\operatorname{div}(\nabla u) = \sum_{i=1}^{d} \partial_{x_i}(\partial_{x_i} u) = \Delta u$$

Preliminaries: divergence formula a.k.a. Stokes theorem

Let Ω be a bounded open set of $\mathbb{R}^d$, with a $\mathcal{C}^1$ boundary. Let $F : \Omega \to \mathbb{R}^d$ be a $\mathcal{C}^1(\overline{\Omega}, \mathbb{R}^d)$ vector field. Then,

$$\int_\Omega \operatorname{div}(F) = \int_{\partial\Omega} F \cdot n \, d\sigma$$

where σ is the surface measure on $\partial\Omega = \overline{\Omega} \setminus \Omega$ and n is the unit outward normal vector to Ω

Applications: $\forall u \in \mathcal{C}^2(\overline{\Omega})$ and $\forall v \in \mathcal{C}^1(\overline{\Omega})$

$$\int_\Omega \Delta u = \int_{\partial\Omega} \partial_n u \, d\sigma \text{ where } \partial_n u = \nabla u \cdot n$$

$$\int_\Omega \nabla u \cdot \nabla v = -\int_\Omega v \Delta u + \int_{\partial\Omega} v \partial_n u \, d\sigma$$

$$\int_\Omega v \Delta u - \int_\Omega u \Delta v = \int_{\partial\Omega} v \partial_n u \, d\sigma - \int_{\partial\Omega} u \partial_n v \, d\sigma$$

Preliminaries: divergence formula a.k.a. Stokes theorem

Examples:

- If $\Omega = (a, b) \subset \mathbb{R}$, $\int_a^b F'(x)\,dx = F(b) - F(a)$
- If $\Omega = (0,1)^2 \subset \mathbb{R}^2$, $\int_\Omega \operatorname{div} F = \int_{\partial\Omega} F \cdot n\, d\sigma$ since
  $\int_0^1 \int_0^1 (\partial_{x_1} F_1 + \partial_{x_2} F_2)(x_1, x_2)\, dx_1\, dx_2 =$
  $\int_0^1 (F_1(1, x_2) - F_1(0, x_2))\, dx_2 + \int_0^1 (F_2(x_1, 1) - F_2(x_1, 0))\, dx_1$
- If $\Omega = B(0, r) \subset \mathbb{R}^2$, then

$$\int_{B(0,r)} \operatorname{div}(F) = \int_{\partial B(0,r)} F \cdot n\, d\sigma = \int_0^{2\pi} F(r\gamma(\theta)) \cdot \gamma(\theta)\, r\, d\theta$$

where $\gamma(\theta) = (\cos(\theta), \sin(\theta))$ is a parametrization of $\partial B(0, 1)$

Derivation of Laplace and Poisson equations

Let us consider a scalar quantity $u(x)$ (temperature, concentration of a chemical species) and the associated flux $F(x)$ (heat flow, material flow)

Step 1: Conservation at equilibrium: in the absence of external sources, the outgoing flux from any domain U is zero: $\forall U \subset \mathbb{R}^d$,

$$\int_{\partial U} F \cdot n \, d\sigma = 0$$

which yields, for all U, $\int_U \operatorname{div}(F) = 0$, and thus

$$\operatorname{div}(F) = 0$$

Derivation of Laplace and Poisson equations

Step 2: Constitutive equation (Fourier law, Fick law): there exists $a > 0$ such that

$$F = -a\nabla u$$

Fluxes create flows from regions with large u to regions with small u, in order to "locally equilibrate" u

Conclusion: one thus obtains (since $\mathrm{div}\,(\nabla u) = \Delta u$)

$$-a\Delta u = 0$$

Remark: If there is a source term f (heater, creation or consumption of matter):

$$\int_{\partial U} F \cdot n \, d\sigma = \int_U f$$

and one obtains

$$-a\Delta u = f$$

Laplace and Poisson equations on domains

In many cases of interest, the equation is posed on an open bounded domain $\Omega \subset \mathbb{R}^d$, with boundary conditions:

Dirichlet problem: fix the value of u on the boundary

$$\begin{cases} -a\Delta u = f \text{ on } \Omega \\ u = g \text{ on } \partial\Omega \end{cases}$$

Neumann problem: fix the out-going flux on the boundary

$$\begin{cases} -a\Delta u = f \text{ on } \Omega \\ \partial_n u = g \text{ on } \partial\Omega \end{cases}$$

Applications

Laplace and Poisson equations appear in numerous situations:

- Stationary thermal diffusion: $-k\Delta T = f$
  (T is the temperature and k is the heat conductivity)
- Stationary diffusion of chemical species: $-D\Delta u = f$
  (u is the concentration and D the diffusivity)
- Electrostatics: $-\Delta V = \frac{\rho}{\epsilon}$
  (V is the electric potential, ρ the charge density and ϵ the permittivity)
- Newtonian gravity: $-\Delta V = -4\pi G\rho$
  (V is the gravitational potential, G the gravitational constant and ρ the mass density)

The Poisson equation in $\mathbb{R}^d$

Find $u : \mathbb{R}^d \to \mathbb{R}$ solution to

$$-\Delta u = f \text{ in } \mathbb{R}^d$$

Two steps:

Step 1: Construction of a solution:

- The fundamental solution of Laplace equation
- A solution to the Poisson problem in $\mathbb{R}^d$ built by convolution against the fundamental solution

We focus for simplicity on $d = 2$: everything can be generalized in dimension $d > 2$ (see the lecture notes)

Step 2: Uniqueness of solutions:

- The mean-value formula for harmonic functions
- Liouville theorem and uniqueness of solutions to the Poisson equation on $\mathbb{R}^d$ (with sublinear growth, up to an additive constant)

The fundamental solution: formula

What are rotation invariant solutions to $-\Delta u = 0$: $u(x) = v(\|x\|)$?
One gets $\nabla u(x) = v'(\|x\|)\frac{x}{\|x\|}$ and thus ($d = 2$):

$$0 = \Delta u(x) = v''(\|x\|) + \frac{v'(\|x\|)}{\|x\|}$$

Solve this ODE to get $v(r) = c_0 \ln r + c_1$, $c_0, c_1 \in \mathbb{R}$

The fundamental solution of $-\Delta$ in $\mathbb{R}^2$ is the function $\Phi : \mathbb{R}^2 \setminus \{0\} \to \mathbb{R}$ defined by:

$$\Phi(x) = -\frac{1}{2\pi} \ln \|x\|$$

For $d \geq 3$, one gets $\Phi(x) = \frac{1}{d(d-2)\omega_d} \frac{1}{\|x\|^{d-2}}$, where ω_d is the volume of the unit ball in $\mathbb{R}^d$

The fundamental solution: properties

- The function $\Phi : \mathbb{R}^2 \setminus \{0\} \to \mathbb{R}$ is $\mathcal{C}^\infty$ and

$$\forall x \in \mathbb{R}^2 \setminus \{0\}, \quad \Delta\Phi(x) = 0$$

- The function Φ goes to ∞ when $\|x\| \to 0$ but

$$\int_{B(0,1)} \Phi(x)\, dx < \infty$$

The function Φ is locally integrable

Indeed $\int_{B(0,1)} \Phi(x)\, dx = -\frac{1}{2\pi} \int_0^1 \ln r \int_0^{2\pi} r\, d\theta dr < \infty$

The fundamental solution: properties

Proposition

For any $\psi \in \mathcal{C}_c^2(\mathbb{R}^d)$ ($\mathcal{C}^2$ functions with compact support),

$$-\int_{\mathbb{R}^d} \Phi \Delta \psi = \psi(0).$$

Proof ($d = 2$):

- $\int_{\mathbb{R}^2} \Phi \Delta \psi = \lim_{\varepsilon \to 0} \int_{B(0,\varepsilon)^c} \Phi \Delta \psi$ (where $B(0,\varepsilon)^c = \mathbb{R}^2 \setminus B(0,\varepsilon)$)
- $\int_{B(0,\varepsilon)^c} \Phi \Delta \psi = \int_{\partial B(0,\varepsilon)} \Phi \partial_n \psi \, d\sigma - \int_{\partial B(0,\varepsilon)} \psi \partial_n \Phi \, d\sigma$
  (where $n = -x/\|x\|$)
- $\int_{\partial B(0,\varepsilon)} \Phi \partial_n \psi \, d\sigma = -\frac{1}{2\pi} \ln \varepsilon \int_0^{2\pi} \partial_n \psi(\varepsilon\gamma(\theta)) \varepsilon d\theta \xrightarrow[\varepsilon \to 0]{} 0$
- $\int_{\partial B(0,\varepsilon)} \psi \partial_n \Phi \, d\sigma = \frac{1}{2\pi\varepsilon} \int_0^{2\pi} \psi(\varepsilon\gamma(\theta)) \varepsilon d\theta \xrightarrow[\varepsilon \to 0]{} \psi(0)$
  since on $\partial B(0,\varepsilon)$, $\partial_n \Phi = -\frac{1}{2\pi} \frac{x}{\|x\|^2} \cdot n(x) = \frac{1}{2\pi\varepsilon}$

Aside on distributions

In modern notation, one writes:

$$-\Delta\Phi = \delta_0$$

in the sense of distributions i.e. for any smooth and compactly supported test function ψ:

$$\langle -\Delta\Phi, \psi \rangle = \langle \delta_0, \psi \rangle$$

By definition, $\langle \delta_0, \psi \rangle := \psi(0)$ and (think of integration by parts)

$$\langle -\Delta\Phi, \psi \rangle := -\int_{\mathbb{R}^d} \Phi \Delta\psi$$

$\rightarrow$ more in the forthcoming lectures "Analyse 2"

Remark (back to physics): Φ is the electric potential created by an electric point charge at $x = 0$ (up to a multiplicative constant)

A solution to the Poisson equation in $\mathbb{R}^d$

Theorem

Let $f \in \mathcal{C}_c^2(\mathbb{R}^d)$, and let us define u by the convolution:

$$u(x) = \int_{\mathbb{R}^d} f(y)\Phi(x-y)\,dy$$

Then $u \in \mathcal{C}^2(\mathbb{R}^d)$ and $-\Delta u = f$ in $\mathbb{R}^d$

Proof: $u(x)$ is well defined and $u(x) = \int_{\mathbb{R}^d} \Phi(y) f(x-y) dy$

- Check that $u \in \mathcal{C}^2(\mathbb{R}^d)$ (dominated convergence) and $-\Delta_x u(x) = -\int_{\mathbb{R}^d} \Phi(y)\Delta_x[f(x-y)]\,dy$
- Since $\Delta_x[f(x-y)] = \Delta_y \psi_x(y)$, where $\psi_x(y) = f(x-y)$, one gets using the previous proposition, for any $x \in \mathbb{R}^d$,

$$-\Delta_x u(x) = -\int_{\mathbb{R}^d} \Phi \Delta \psi_x = \psi_x(0) = f(x)$$

Uniqueness?

We identified *one* solution to the Poisson equation $-\Delta u = f$ in $\mathbb{R}^d$. Is it the only one?

Assume that $-\Delta v = f$ is another solution. Then

$$\Delta(u - v) = 0 \text{ in } \mathbb{R}^d$$

To understand uniqueness, we thus need to identify the harmonic functions in $\mathbb{R}^d$

Two remarks:

- Of course, constants are harmonic, and thus $v = u + C$ is a solution: to get a unique solution, one needs at least to impose a condition to determine C (average of u, value of u at one point,...)
- But $a \cdot x$ or $x_1^2 - x_2^2$ are also harmonic functions...

The mean-value formula

Theorem

Let $u \in \mathcal{C}^2(U)$ (U open set of $\mathbb{R}^d$) be such that $-\Delta u = 0$ in U
Then for any $x \in U$ and any $r > 0$ (such that $\overline{B(x,r)} \subset U$)

$$u(x) = \fint_{\partial B(x,r)} u \, d\sigma = \fint_{B(x,r)} u$$

where $\fint_{\partial B(x,r)} = \frac{1}{|\partial B(x,r)|} \int_{\partial B(x,r)}$ and likewise $\fint_{B(x,r)} = \frac{1}{|B(x,r)|} \int_{B(x,r)}$

Proof ($d = 2$) Let, for $r > 0$,

$$m(r) = \fint_{\partial B(x,r)} u \, d\sigma = \frac{1}{2\pi r} \int_0^{2\pi} u(x + r\gamma(\theta)) \, r d\theta$$

One has, thanks to divergence formula

$$m'(r) = \frac{1}{2\pi} \int_0^{2\pi} \nabla u(x + r\gamma(\theta)) \cdot \gamma(\theta) \, d\theta = \fint_{\partial B(x,r)} \partial_n u \, d\sigma = 0$$

Moreover, $\int_{B(x,r)} u = \int_0^r \int_0^{2\pi} u(x + \rho\gamma(\theta)) \rho d\theta d\rho = u(x) \int_0^r 2\pi\rho \, d\rho$

Liouville theorem

Theorem

Assume that $u \in \mathcal{C}^2(\mathbb{R}^d)$ is harmonic and s.t. $|u(x)| = o(|x|)$ (when $x \to \infty$):

$$\forall \varepsilon > 0,\ \exists M > 0,\ \forall x \in \mathbb{R}^d, |x| > M \Rightarrow |u(x)| \leq \varepsilon |x|$$

Then u is constant

Proof ($d = 2$) If u is harmonic, so is $\partial_{x_i} u$ for any $i \in \{1, \dots, d\}$

$$\begin{aligned}
|\partial_{x_i} u(x_0)| &= \left| \fint_{B(x_0,r)} \partial_{x_i} u \right| = \frac{1}{\pi r^2} \left| \int_{B(x_0,r)} \operatorname{div}(u e_i) \right| \\
&= \frac{1}{\pi r^2} \left| \int_{\partial B(x_0,r)} u\, n_i \, d\sigma \right| \leq \frac{1}{\pi r^2} \int_{\partial B(x_0,r)} |u| d\sigma \\
&\leq \frac{2\pi r}{\pi r^2} \|u\|_{L^\infty(\partial B(x_0,r))} = \frac{2}{r} \|u\|_{L^\infty(\partial B(x_0,r))} = \frac{2}{r} o(r) \xrightarrow[r \to \infty]{} 0
\end{aligned}$$

Corollary of Liouville theorem: a uniqueness result

Corollary

Let $f \in \mathcal{C}_c^2(\mathbb{R}^d)$. Let u be a solution of $-\Delta u = f$ in $\mathbb{R}^d$ with sublinear growth at infinity ($|u(x)| = o(|x|)$). Then, there exists a constant $C \in \mathbb{R}$ such that for all $x \in \mathbb{R}^d$

$$u(x) = \int_{\mathbb{R}^d} \Phi(x-y) f(y)\, dy + C$$

Proof: One can check that $\tilde{u}(x) = \int_{\mathbb{R}^d} \Phi(x-y) f(y)\, dy$ has sublinear growth at infinity
Thus, if u is another solution s.t. $|u(x)| = o(|x|)$, then $u - \tilde{u}$ is harmonic with sublinear growth at infinity, and thus constant

The Dirichlet problem

Let us now consider the Dirichlet problem:

$$\begin{cases} -\Delta u = f \text{ on } \Omega \\ \quad u = g \text{ on } \partial\Omega \end{cases}$$

on a bounded open set $\Omega \subset \mathbb{R}^d$

- How to build a solution? $\rightarrow$ A representation formula using Green's function
- Is it unique? $\rightarrow$ Maximum principle and a uniqueness result

A representation formula for the Dirichlet problem

The Green's function for the Dirichlet problem on Ω is defined by: for all $x, y \in \Omega$, with $x \neq y$,

$$G(x, y) = \Phi(y - x) - h_x(y)$$

where the corrector function h_x is solution to

$$\begin{cases} -\Delta_y h_x(y) = 0 \text{ for all } y \in \Omega, \\ h_x(y) = \Phi(y - x) \text{ for all } y \in \partial\Omega. \end{cases}$$

Theorem

Let $u \in \mathcal{C}^2(\overline{\Omega})$ be a solution to the Dirichlet problem. Then

$$u(x) = \int_\Omega G(x, y) f(y)\, dy - \int_{\partial\Omega} \partial_n G(x, y) g(y) \sigma(dy)$$

where $\partial_n G(x, y) = \nabla_y G(x, y) \cdot n(y)$ is the outward normal derivative of G with respect to the y variable.

A representation formula for the Dirichlet problem

Proof: (Recall that $-\Delta_y \Phi(y-x) = \delta_x$ and $-\Delta_y h_x(y) = 0$)

$$\begin{aligned}\int_\Omega G(x,y) f(y)\,dy &= -\int_\Omega \Phi(y-x)\Delta u(y)\,dy + \int_\Omega h_x(y)\Delta u(y)\,dy \\ &= u(x) - \int_{\partial\Omega} \Phi(y-x)\partial_n u(y)\,\sigma(dy) + \int_{\partial\Omega} u(y)\partial_n \Phi(y-x)\,\sigma(dy) \\ &\quad + \int_{\partial\Omega} h_x(y)\partial_n u(y)\,\sigma(dy) - \int_{\partial\Omega} \partial_n h_x(y) u(y)\,\sigma(dy) \\ &= u(x) + \int_{\partial\Omega} \partial_n G(x,y) g(y)\,\sigma(dy)\end{aligned}$$

Remark: This is a representation formula: one needs some (nontrivial) assumptions on f and g to turn it into an existence result

Remark: Building h_x in practice is complicated...

$\rightarrow$ This is not a very robust approach: see "Equations aux dérivées partielles : approches variationnelles" for alternatives!

A uniqueness result for the Dirichlet problem

A very powerful corollary of the mean value formula is the maximum principle

Theorem

Assume that $u \in \mathcal{C}^2(U) \cap \mathcal{C}(\overline{U})$ is harmonic in U, open and bounded subset of $\mathbb{R}^d$. The weak maximum principle states that

$$\max_{\overline{U}} u = \max_{\partial U} u$$

The strong maximum principle states that, if in addition U is connected and $\exists x_0 \in U, u(x_0) = \max_{\overline{U}} u$, then u is constant in U

Proof: If $x_0 \in U$, $u(x_0) = \max_{\overline{U}} u$, then, for any $r > 0$ s.t. $B(x_0, r) \subset U$,

$$u(x_0) = \fint_{B(x_0, r)} u$$

which implies $u = u(x_0)$ in $B(x_0, r)$. Thus $\{x \in U : u(x) = u(x_0)\}$ is open, and also closed and thus $u = u(x_0)$ on the connected component of U which contains x_0

A uniqueness result for the Dirichlet problem

Corollary

There exists at most one solution of the Dirichlet problem with regularity $\mathcal{C}^2(\Omega) \cap \mathcal{C}(\overline{\Omega})$

Proof: If u and v are solutions to the Dirichlet problem, then $w = u - v$ satisfies:

$$\begin{cases} -\Delta w = 0 \text{ on } \Omega \\ \quad w = 0 \text{ on } \partial\Omega \end{cases}$$

Thus, by the weak maximum principle $\max_{\overline{\Omega}} w = \max_{\partial\Omega} w = 0$ which yields

$$w \leq 0$$

Likewise $\max_{\overline{\Omega}}(-w) = \max_{\partial\Omega}(-w) = 0$ which yields

$$-w \leq 0$$

Thus $w = 0$

Discretization using finite differences

Let us consider the Dirichlet problem in dimension 1 ($\Omega = (0,1)$):

$$\begin{cases} -u''(x) = f(x) \text{ for } x \in (0,1) \\ u(0) = a \text{ and } u(1) = b \end{cases}$$

Assume $f \in \mathcal{C}^0(\overline{\Omega})$ so that there exists a unique solution $u \in \mathcal{C}^2(\Omega) \cap \mathcal{C}^0(\overline{\Omega})$: how to get a numerical approximation of u?

Remark: We stick to dimension $d = 1$ to keep things simple: everything can be generalized to higher dimensions (in practice, if $d \geq 5$, the algorithm is too costly...) See the lecture notes for supplementary materials

The finite difference method

Step 1: Introduce a discretization of $(0,1)$ into I intervals: $(x_i)_{0\leq i\leq I}$ where $x_i = ih$ and $h = \frac{1}{I}$

Step 2: We already know $u(x_0) = a$ and $u(x_I) = b$. Write the PDE at the interior points:

$$-u''(x_i) = f(x_i) \text{ for } i \in \{1, \dots, I-1\}$$

Step 3: Approximate the differential operator using finite differences: assuming $u \in \mathcal{C}^4(\overline{\Omega})$, for some $h_+, h_- \in (0,h)$,

$$u(x_{i+1}) = u(x_i) + hu'(x_i) + \frac{h^2}{2}u''(x_i) + \frac{h^3}{6}u^{(3)}(x_i) + \frac{h^4}{24}u^{(4)}(x_i + h_+)$$

$$u(x_{i-1}) = u(x_i) - hu'(x_i) + \frac{h^2}{2}u''(x_i) - \frac{h^3}{6}u^{(3)}(x_i) + \frac{h^4}{24}u^{(4)}(x_i - h_-).$$

By adding these two equalities, one gets

$$u''(x_i) = D_h^2 u(x_i) + O(h^2) \text{ where } D_h^2 u(x_i) = \frac{u(x_{i+1}) - 2u(x_i) + u(x_{i-1})}{h^2}$$

The discretized problem

One thus gets the discretized problem: find $U \in \mathbb{R}^{I-1}$ such that, for all $i \in \{1, \dots, I-1\}$,

$$-\frac{U_{i+1} - 2U_i + U_{i-1}}{h^2} = F_i$$

where $F_i = f(x_i)$, $U_0 = a$ and $U_I = b$

This is a linear problem: $AU = C + F$ ($A \in \mathbb{R}^{(I-1)\times(I-1)}$; $C, F \in \mathbb{R}^{I-1}$)

$$A = \frac{1}{h^2}\begin{bmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & -1 & 2 & -1 \\ 0 & \cdots & 0 & -1 & 2 \end{bmatrix} \quad C = \frac{1}{h^2}\begin{pmatrix} a \\ 0 \\ \vdots \\ 0 \\ b \end{pmatrix} \quad \text{and} \quad F = \begin{pmatrix} f(x_1) \\ f(x_2) \\ \vdots \\ f(x_{I-2}) \\ f(x_{I-1}) \end{pmatrix}$$

The discretized problem

The matrix A is symmetric positive definite and thus there exists a unique U solution to $AU = C + F$

Indeed, for $V \in \mathbb{R}^{I-1}$, $V^T AV = \sum_{i=0}^{I-1} \left(\frac{V_{i+1} - V_i}{h} \right)^2$ with convention $V_0 = V_I = 0$: this is the discrete analogue of $-\int_0^1 u'' u = \int_0^1 (u')^2$

Let us introduce the error vector

$$E = \pi(u) - U$$

where $\pi(u) \in \mathbb{R}^{I-1}$ is defined by: $\forall i \in \{1, \dots, I-1\}$,

$$[\pi(u)]_i = u(x_i)$$

We would like to study the asymptotics of $\|E\|$ in the limit $h \to 0$, for some vector norm $\|\cdot\|$

Convergence analysis

The convergence analysis relies on two steps:

- A consistency analysis: what are the errors when one plugs the exact solution in the numerical scheme?
- A stability analysis: how do the consistency errors propagate to the global error $\|E\|$?

We will consider two norms in the following:

- The l^∞-norm: $\|E\|_\infty = \sup_{1 \leq i \leq l-1} |E_i|$
- The l^2-norm: $\|E\|_2 = \sqrt{\frac{1}{l-1} \sum_{i=1}^{l-1} |E_i|^2}$

Remark: Why this prefactor $\frac{1}{l-1}$?
(i) $\|E\|_2 \leq \|E\|_\infty$; (ii) $\lim_{h \to 0} \|\pi(u)\|_2^2 = \int_0^1 u^2(x)\, dx$

Consistency error

The consistency error vector ε is defined by:

$$A\pi(u) = C + F + \varepsilon$$

i.e. for any $i \in \{1, \dots, I-1\}$,

$$-\frac{u(x_{i+1}) - 2u(x_i) + u(x_{i-1})}{h^2} = f(x_i) + \varepsilon_i$$

Since $f(x_i) = -u''(x_i)$, one thus gets:

$$\varepsilon_i = u''(x_i) - D_h^2 u(x_i)$$

and thus, if $u \in \mathcal{C}^4(\overline{\Omega})$, there exists C such that for all h small enough,

$$\|\varepsilon\|_\infty \leq Ch^2$$

Stability analysis

We have on the one hand

$$AU = C + F$$

and on the other hand

$$A\pi(u) = C + F + \varepsilon$$

thus (recall that $E = \pi(u) - U$)

$$AE = \varepsilon$$

Question: if $\|\varepsilon\|_\infty \leq Ch^2$, do we have $\|E\|_\infty \leq Ch^2$?

Difficulty: when $h \to 0$, $I = 1/h \to \infty$: we have more and more terms to control...

Aside on induced matrix norms

Definition

Let $\|\cdot\|$ be a vector norm on $\mathbb{R}^n$ and let $M \in \mathbb{R}^{n\times n}$ be a square matrix. Then the induced matrix norm is defined by:

$$\|M\| = \sup_{x\in\mathbb{R}^n, \|x\|\leq 1} \|Mx\|$$

Standard properties: for $M, N \in \mathbb{R}^{n\times n}$ and $x \in \mathbb{R}^n$,

$$\|Mx\| \leq \|M\|\|x\|$$

$$\|MN\| \leq \|M\|\|N\|$$

Example: For the l^2-norm,

$$\|M\|_2^2 = \sup_{x\in\mathbb{R}^n, \|x\|\leq 1} (Mx)\cdot(Mx) = \sup_{x\in\mathbb{R}^n, \|x\|\leq 1} x^T M^T Mx = \rho(M^T M)$$

where ρ is the spectral radius: $\rho(M^T M) = \max_{1\leq i\leq n} |\lambda_i|$ where $(\lambda_1, \ldots, \lambda_n)$ are the eigenvalues of $M^T M$

Back to the stability analysis

We have $\|\varepsilon\|_\infty \leq Ch^2$ and $AE = \varepsilon$.
Thus

$$\|E\|_\infty \leq \|A^{-1}\|_\infty \|\varepsilon\|_\infty \leq C\|A^{-1}\|_\infty h^2$$

and

$$\|E\|_2 \leq \|A^{-1}\|_2 \|\varepsilon\|_2 \leq C\|A^{-1}\|_2 h^2$$

The numerical scheme is stable in l^∞-norm (resp. l^2-norm) if $\|A^{-1}\|_\infty \leq C$ (resp. $\|A^{-1}\|_2 \leq C$) with C independent of h

If this is the case, the scheme is convergent with order 2 in l^∞-norm (resp. l^2-norm)

Lax principle: consistency + stability $\Rightarrow$ convergence

Stability in l^2-norm

Remember that A is symmetric positive definite. We need to analyze

$$\|A^{-1}\|_2 = \sqrt{\rho(A^{-T}A^{-1})} = \sqrt{\rho(A^{-2})} = \rho(A^{-1})$$
$$= \frac{1}{\min(\lambda, \lambda \text{ eigenvalue of } A)}$$

We would like to show that $\|A^{-1}\|_2 \leq C$ and thus

$$\min(\lambda, \lambda \text{ eigenvalue of } A) \geq 1/C$$

($1/C$ independent of h)

Stability in l^2-norm

Lemma

Let us consider $\tilde{A} \in \mathbb{R}^{(I-1)\times(I-1)}$*:*

$$\tilde{A} = \begin{bmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & -1 & 2 & -1 \\ 0 & \cdots & 0 & -1 & 2 \end{bmatrix}$$

The eigenvalues of $\tilde{A}$ *are* $\lambda_i = 4\sin^2\left(\frac{i\pi}{2I}\right)$ $(1 \leq i \leq I-1)$ *with eigenvectors* $v_i \in \mathbb{R}^{I-1}$ *defined by*

$$\forall j \in \{1, \ldots, I-1\}, [v_i]_j = \sin\left(\frac{ij\pi}{I}\right)$$

Stability in l^2-norm

Since $A = \frac{1}{h^2}\tilde{A}$, one gets

$$\min(\lambda, \lambda \text{ eigenvalue of } A) = \frac{1}{h^2}\lambda_1 = \frac{4}{h^2}\sin^2\left(\frac{\pi h}{2}\right) \geq C$$

since $\sin(x) \geq Cx$ for $x \in [0, \pi/2]$

This shows that the finite difference scheme is stable in l^2-norm

Stability in l^∞-norm

We would like to estimate $\|A^{-1}\|_\infty$. Two ingredients: (i) a discrete maximum principle and (ii) a barrier vector

(i) The numerical scheme satisfies a discrete maximum principle: if $AV = F$ and $F \geq 0$ then $V \geq 0$

(ii) Let $\varphi \in \mathbb{R}^{I-1}$ be the vector s.t. $\varphi_i = \frac{x_i(1-x_i)}{2}$. Then $A\varphi = 1$

Recall that A is a discretization of $-\partial_{x,x}$ (with Dirichlet boundary conditions): (i) is a discrete counterpart of the weak maximum principle; (ii) is a discrete counterpart of $-\partial_{x,x}\Big(x(1-x)/2\Big) = 1$

Consequence: For all $h > 0$, $\|A^{-1}\|_\infty \leq 1/8$. The finite difference scheme is stable in l^∞-norm

Proof: Let $AU = F$: we would like to prove that $\|U\|_\infty \leq \|F\|_\infty/8$
$A(\|F\|_\infty\varphi - U) = \|F\|_\infty - F \geq 0$ and thus $\|F\|_\infty\varphi - U \geq 0$
$A(\|F\|_\infty\varphi + U) = \|F\|_\infty + F \geq 0$ and thus $\|F\|_\infty\varphi + U \geq 0$
Therefore $\|U\|_\infty \leq \|F\|_\infty\|\varphi\|_\infty = \|F\|_\infty/8$

Convergence results

Theorem

Let us consider the one-dimensional Dirichlet problem with solution $u : (0,1) \to \mathbb{R}$ *and the associated finite difference approximation with solution* $U \in \mathbb{R}^{I-1}$. *Let us assume that* $u \in \mathcal{C}^4([0,1])$. *Then, there exists* $C > 0$ *such that, for any* $h > 0$,

$$\|U - \pi(u)\|_2 \leq \|U - \pi(u)\|_\infty \leq Ch^2$$

- If $u \notin C^4([0,1])$, the convergence rate may be affected: the convergence rate is limited by (i) the order of the numerical scheme and (ii) the regularity of the solutions
- The finite difference method is of interest but (i) it is restricted to specific domains in dimension $d > 1$; (ii) the convergence analysis is cumbersome. $\to$ See forthcoming lectures on Galerkin methods (finite element, spectral)
- The complexity of all these techniques scales badly with the dimension (curse of dimensionality)